



Rating
Buy

North America
United States

Industrials
Space Technology

Company
SpaceX

Reuters
SPCX.OO

Bloomberg
SPCX US

Exchange
NSM

Ticker
SPCX

Date
14 July 2026

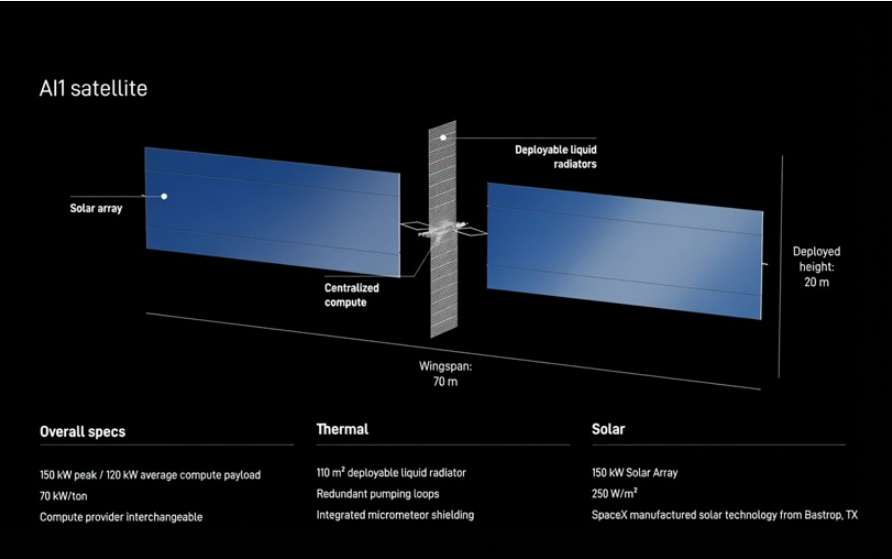
Special Report

Price at 13 Jul 2026 (USD)	139.14
Price Target	255.00
52-week range	201.80 - 139.14

Data Centers in Space Part 4

Following up on our [initiation](#) last week, we take a closer look at orbital data centers and continue our *Data Centers in Space* series, introducing a refreshed cost model (Excel available upon request) and further analysis around optical/lasers, spectrum, solar, radiator, and compute density. We factor in the latest disclosures from SpaceX regarding its AI1 satellite and Starlink constellation. Our high-level conclusion is the physics checks out and the biggest challenge will be cost/scale, which we think SpaceX can achieve through extreme vertical integration. Combined with Starship launches scaling up, we envision a scenario in the early 2030s where cost can be on par with terrestrial.

Figure 1: SpaceX AI1 satellite design



Source : Company website

Valuation & Risks

Edison Yu
Research Analyst
+1-212-250-7263

Bryan Kraft
Research Analyst
+1-212-250-0117

Corinne Blanchard
Research Analyst
+1-904-645-2360

Gary Zhou, CFA
Research Analyst
+852-2203-5889

Ross Seymore
Research Analyst
+1-415-617-3268

Roshan Ranjit, CFA
Research Analyst
+44-20-754-52908

Laura Li
Research Associate
+1-212-250-2266

Deutsche Bank Securities Inc.

IMPORTANT RESEARCH DISCLOSURES AND ANALYST CERTIFICATIONS LOCATED IN APPENDIX 1. Deutsche Bank does and seeks to do business with companies covered in its research reports. Thus, investors should be aware that the firm may have a conflict of interest that could affect the objectivity of this report. Investors should consider this report as only a single factor in making their investment decision.



Optical lasers & spectrum

Starmind

The AI1 satellites will rely on optical inter-satellite links (OISLs) to route traffic within the Starmind constellation and then into the Starlink constellation, whose laser mesh network then backhauls traffic to ground stations. As such, AI1 satellites do not require a complex multi-panel phased-array antenna onboard (i.e., no meaningful spectrum needed for communication except for possibly some Ka-band as a backup for telemetry). We estimate there are >10,000 Starlink satellites in orbit now with V2 minis each having 3 optical terminals capable of handling ~600 Gbps of capacity (200 Gbps per terminal). For the upcoming V3 satellites, we suspect there may be at least 2-3 Tbps of capacity per satellite. With the quantity of data increasing, we also anticipate SpaceX augmenting and/or upgrading its ground station network.

Figure 2: Starlink mini laser terminal



Source : Company reports

Leading providers of optical terminals are Tesat-Spacecom (subsidiary of Airbus), Mynaric (acquired earlier this year), SA Photonics (acquired by CACI in 2021) and SpaceX (uses internally but also started selling “mini” laser terminals externally - Muon Space and Starcloud have announced plans to use).

Starlink

Using such an architecture, this means the space-to-ground layer will bear the burden of a lot more traffic. Optical links make the Starmind constellation itself spectrum-free, but data that leaves the laser mesh network must ultimately funnel through Starlink's gateways, and orbital data centers change the shape of that data flow. Starlink's Gen1/Gen2 gateway authorizations were sized for a consumer broadband business whose traffic is asymmetrical (downlink heavy). As such, SpaceX is increasingly utilizing more unconventional spectrum bands beyond Ku-/Ka- such as:

- **E-band:** already using for gateway backhaul; uplink+downlink
- **V-band:** authorized for advanced user terminals and gateway; uplink +downlink
- **W-band:** authorized for gateway backhaul; primarily uplink



- **D-band:** proposed for gateway backhaul mainly for AI; primarily uplink

Higher bands historically face severe rain fade, oxygen absorption, and atmospheric attenuation that make them unreliable for consumer user terminals so it appears Starlink is mostly using them for high-capacity gateway backhaul links (where large dishes, site diversity, and optical inter-satellite routing can mitigate these issues), prioritizing proven Ku-/Ka- for consumer antennas while gradually adding V- for advanced users as hardware and regulations mature.

Separately, the FCC recently adopted new standards for satellites, replacing the 1990s-era Equivalent Power Flux Density (EPFD) limits that forced LEO systems like Starlink to severely restrict power and beam density to protect GEO satellites. Instead, the new rules use performance-based protection criteria that account for today's adaptive coding/modulation, beamforming, and real-world coordination. In the key Ku-/Ka- bands (10.7–12.7 GHz, 17.3–18.6 GHz, 19.7–20.2 GHz), this allows operators to run at higher power and many more co-frequency satellites/beams serving the same area (e.g., 7–8 instead of effectively 1). The FCC cited up to 7× more capacity from the same number of satellites.

Solar power

Current solutions

Satellites mainly use **multi-junction** or **silicon** solar cells. Multi-junction delivers higher performance (~30% or higher efficiency) by stacking different materials such as gallium arsenide (GaAs) and germanium (Ge) which enable broader spectrum capture of sunlight; these materials also provide greater radiation resistance. However, the production of these two materials is highly concentrated in China especially gallium which can be produced as a byproduct aluminum refining and similarly for germanium, as a byproduct of zinc smelting. Spectrolab (subsidiary of Boeing), AZUR Space (based in Germany; acquired by 5N Plus in 2021), and SolAero (acquired in 2022) are leading providers of multi-junction space grade cells. Silicon is materially cheaper (up to 80%) benefitting from much larger industrial scale but performs worse (usually 15-20% efficiency), and is not as durable due to lower radiation resistance. Within silicon, the main type of cell is referred to as a passivated emitter and rear cell (PERC). Taiwan Solar Energy Corp (TSEC) is a supplier of this type of cell to SpaceX. Amazon LEO also uses PERC but it is unclear who the supplier is.

What's coming next?

In the next few years, we think a new type of approach can emerge for satellites: **heterojunction** (HJT). This type of cell combines crystalline silicon with thin layers of amorphous silicon to improve performance. Interestingly, HJT uses a simpler (5-7 steps vs. typical 12-13), lower-temperature production process that could save up to 70% on energy usage although upfront equipment costs will be higher; there is also potential to remove silver from the coating and use copper instead, saving cost. Performance wise, efficiency has been recorded to be as high as 27% and the symmetrical design allows the cell to absorb almost equal light from both sides. Lastly, HJT is very radiation hardened and degrades slowly in orbit. In the US, Solestial (recently acquired by York Space Systems) is working on development of ultrathin, flexible silicon HJT solar cells, leveraging equipment from Meyer Burger.

Beyond HJT, the use of **perovskite** tandem cells may be the future. Perovskite is a synthetic crystal structure that essentially can be printed or sprayed onto a surface, meaning it is very lightweight and theoretically as efficient as multi-junction. Solar



cells today are all rigid but with perovskite, the ability to utilize thin-film or flexible panels becomes possible. Rigid (whether multi-junction or silicon) means the cells are mounted onto a honeycomb-like substrate made of aluminum or carbon fiber composite. These panels then are stacked and use mechanical hinges and springs to unfold once in orbit.

In China, Drinda (2865.HK) showcased a dedicated “space PV” zone at [SNEC](#) in May, highlighting its efforts to develop perovskite-based solar solutions for space applications. The company displayed its perovskite/TOPCon tandem technology (with ~33.5% small-area efficiency), alongside flexible CPI-based substrates and a commercial satellite model. The company has been an early mover in the space-based solar business. Xuntian, a subsidiary of the company, plans to manufacture and deploy 12-15 satellites in 2026 (with an order backlog of 54 satellites). GCL’s (3800.HK) booth design and product display at SNEC reflected a clear shift in focus beyond traditional PV, with both battery materials (cathode and anode) and space solar prominently positioned at the front of its exhibition area. For space solar, the company showcased a flexible perovskite solar cell sample alongside satellite-related use cases.

SpaceX roadmap

SpaceX is currently building a solar cell manufacturing facility in Bastrop, Texas near Austin, as part of a much larger ambition for domestic solar cell production. Bastrop is targeting 10 GW in solar cell capacity with two floors (5 GW each). The facility is permitted for ~1.1m square feet, with potential expansion to 1.6 million, and is co-located with SpaceX’s existing Starlink production site. Construction on the solar factory began in late March 2026, with equipment installation having already begun. The plant aims for production ramp-up and “reasonable volume” by the end of 2027. Our understanding is that the plant will initially produce silicon solar cells, delivering around 19% efficiency.

Figure 3: SpaceX ODC satellite production site



Source : Company website

Looking ahead, SpaceX may switch to HJT or perovskite cells but we suspect this will not occur in the near-term option as they likely can not secure the appropriate equipment in time for the AI1 satellite deployment. Ultimately, Elon Musk has indicated plans to put up 100 GW of domestic solar cell capacity within three years.



Radiator

Thermal management

Satellites operate in the vacuum of space, meaning there's no air or medium for heat to dissipate via conduction or convection like on Earth. Instead, satellites rely on thermal radiation to reject excess heat generated by onboard electronics and other components. Radiators function by emitting electromagnetic energy, primarily in the infrared spectrum, which carries away heat as photons. The process follows the Stefan-Boltzmann law, where the rate of heat radiation is dependent on temperature and surface area.

Passive or active?

There are generally two types of radiators: **passive** and **active**. Passive radiators reject heat using nothing but material properties and geometry. This exploits the intrinsic thermal properties of materials, surfaces, and geometric configurations to manage heat flow without consuming any electrical power. The main drawback is that the radiator area is limited by the spacecraft's form factor, which caps passive heat rejection capacity. Active radiators are part of a powered thermal loop — pumps circulate fluid from cold plates on the electronics to the radiator panel. This costs power and adds failure modes, but handles much higher heat loads and holds tighter temperature control than passive designs. Nearly all satellites use passive radiator(s). Active radiators are essentially only used on space stations (ISS, Tiangong) and capsules (Dragon, Starliner).

Historically, primes built most of their own radiator panels in-house. Merchant suppliers include Advanced Cooling Technologies (ACT), Paragon Space Development, Redwire, Beyond Gravity, Iberespacio, and Admatis.

SpaceX roadmap

SpaceX produces the radiator on current Starlink satellites in-house (passive) and will continue to do so for AI1 satellite (active). AI1 will use a double-sided active deployable liquid radiator capable of dissipating $1,400 \text{ W/m}^2$ (700 W/m^2 per face) covering a 110 m^2 surface area. This type of radiator requires electrical power and/or moving parts to transport heat from the compute module to the radiator panels. This is necessary given the high power 120–150 kW payload, where heat fluxes are too high for purely passive conduction. Active radiators have typically been very low volume components so SpaceX will be the first to mass produce such a design.

Compute density

SpaceX is targeting prototype deployments of its AI1 satellites late next year. FCC-filings outline a constellation of up to one million satellites in LEO. Because each satellite has a fixed power and heat budget, the imperative is to maximize computational density. SpaceX is targeting density of 100 kW per ton but AI1 will not meet this threshold, starting off at ~70 kW, implying about 6 MW of compute per Starship launch initially (assumed 85 ton payload capacity). SpaceX expressed openness to hosting all types of chip payloads for its AI1 satellites including Nvidia GPUs, Google TPUs, Amazon Trainium, and of course Tesla's AI chips which prioritize energy efficiency. AI1 is expected to run sustainably at about 120 kW, roughly in line with the power consumption of a NVIDIA GB300 NVL72 rack. In our model, we assume compute density improves to closer to 85-90 kW in 2032E and then 100 kW with the AI3 satellite beyond that.



Economics can eventually be compelling

To set a baseline, we assume upfront capex for 1 GW of AI compute on the ground is \$38bn and requires \$900m in annual opex (power, maintenance, labor, etc...), translating into \$42.5bn over 5 years (based on Epoch AI [analysis](#)). Of this amount, compute represents \$21bn and we assume this carries over to orbital data centers at a 10% mark-up to account for overprovisioning in case of GPU failures. Our sense is GPU failures will not be addressed directly by maintenance but simply each satellite will just operate at lower power in the event of a failure. Therefore, the non-compute costs for terrestrial are \$21.5bn and ODC essentially has to break below this level post overprovisioning or \$19.5bn to be at parity. Looking forward, we assume the cost of terrestrial increases going forward. To illustrate, NVIDIA's CEO Jensen Huang recently commented at GTC Taipei 2026 that a new 1 GW "AI factory" could approach \$100bn in cost with roughly half being compute-related.

Using current SpaceX launch vehicles and satellite designs, we estimate the near-term cost of deploying a 1 GW space data center constellation would be 6x higher than terrestrial (ex-compute). This gap can narrow to 1.0-1.5x later by end of the decade and then eventually be cheaper in the early-mid 2030s. This reduction is primarily driven by Starship (rapid reusability) and aggressive optimization/scaling of the AI-series satellites. *We also refresh the DB Orbital Data Center Model which goes deeper into the economic viability of launching AI infrastructure into orbit (Excel available upon request).*

Figure 4: Terrestrial vs. orbital data center deployment cost

	2027 (generic)	2029 (AI1)	2032 (AI2)	Beyond (AI3)
Launch cost per kg	\$1,429	\$398	\$170	\$43
Satellite cost ex compute (\$/kW)	50,133	13,256	8,857	5,008
Compute density (W/kg)	33	56	87	98
Orbital cost (\$bn)	115	27	15	9
Terrestrial cost (ex compute)	20	23	25	25
Cost difference factor	5.9x	1.2x	0.6x	0.3x

Source : Company reports, Deutsche Bank Research

Figure 5: Starship launch cost trajectory assumptions

(\$mn)	BOM	Variable	Fixed	Total	Payload (tons)	Price/kg
Starship (early, no ship reuse)	52	2	267	321	65	\$4,933
Starship (partial reuse)	5	2	27	34	85	\$398
Starship (full reuse)	2	2	13	17	100	\$170
Starship (full+rapid)	1	1	4	6	200	\$32

Source : Company reports, Deutsche Bank Research



Figure 6: Orbital data center satellite cost trajectory assumptions

(\$000)	2027 (generic)	2029 (AI1)	2032 (AI2)	Beyond (AI3)
Solar	1,750	600	375	300
Radiator	709	438	204	127
Compute	1,620	2,880	1,440	720
Bus structure	500	600	500	400
Laser	300	150	100	75
Other	250	200	150	100
Total	5,129	4,868	2,769	1,722
Ex compute	3,509	1,988	1,329	1,002
Satellite size (kW)	70	150	150	200
Satellite compute (kW)	50	120	130	195
PUE	1.40	1.25	1.15	1.03
Satellite hardware cost (\$/W)	73	32	18	9
Ex compute (\$/W)	50	13	9	5
Mass (kg)	1,500	2,140	1,500	2,000
Power density (W/kg)	47	70	100	100
Compute density (W/kg)	33	56	87	98
Solar cell efficiency	20%	19%	24%	30%
Solar Constant (AM0)	1,366	1,366	1,366	1,366
Degradation + pointing margin	5%	5%	5%	5%
Solar area (m ²)	270	600	482	514
Solar performance (W/m ²)	260	250	311	389
\$/W	\$25.00	\$4.00	\$2.50	\$1.50
\$/m ²	\$6,489	\$999	\$779	\$584
Radiator area (m ²)	89	110	102	127
Radiator heat rejection (W/m ²)	750	1,300	1,400	1,500
\$/m ²	\$8,000	\$4,000	\$2,000	\$1,000
Cost per chip (\$)	\$45,000	\$40,000	\$20,000	\$10,000
# of chips (GB300e)	36	72	72	72

Source : Company reports, Deutsche Bank Research



Appendix 1

Important Disclosures

*Other information available upon request

Disclosure checklist			
Company	Ticker	Recent price*	Disclosure
SpaceX	SPCX.OQ	139.14 (USD) 13 Jul 2026	1, 7, 8, 14, 15, 24, 26

*Prices are current as of the end of the previous trading session unless otherwise indicated and are sourced from local exchanges via Reuters, Bloomberg and other vendors. Other information is sourced from Deutsche Bank, subject companies, and other sources. For disclosures pertaining to recommendations or estimates made on securities other than the primary subject of this research, please see the most recently published company report or visit our global disclosure look-up page on our website at <https://research.db.com/Research/Disclosures/EquityResearchDisclosures>. Aside from within this report, important risk and conflict disclosures can also be found at <https://research.db.com/Research/Disclosures/Disclaimer>. Investors are strongly encouraged to review this information before investing.

Important Disclosures Required by U.S. Regulators

Disclosures marked with an asterisk may also be required by at least one jurisdiction in addition to the United States. See Important Disclosures Required by Non-US Regulators and Explanatory Notes.

1. Within the past year, Deutsche Bank and/or its affiliate(s) has managed or co-managed a public offering for this company, for which it received fees.
7. Deutsche Bank and/or its affiliate(s) has received compensation from this company for the provision of investment banking or financial advisory services within the past year.
8. Deutsche Bank and/or its affiliate(s) expects to receive, or intends to seek, compensation for investment banking services from this company in the next three months.
14. Deutsche Bank and/or its affiliate(s) has received compensation from this company within the past year for non-investment banking related services.
15. This company has been a client of Deutsche Bank Securities Inc. within the past year during which time it received investment banking services.

Important Disclosures Required by Non-U.S. Regulators

Disclosures marked with an asterisk may also be required by at least one jurisdiction in addition to the United States. See Important Disclosures Required by Non-US Regulators and Explanatory Notes.

1. Within the past year, Deutsche Bank and/or its affiliate(s) has managed or co-managed a public offering for this company, for which it received fees.
24. Deutsche Bank and/or its affiliate(s) is or has been over the previous 12 months party to an agreement with the company relating to the provision of services set out in Sections A and B of Annex I of Directive 2014/65/EU, or has over the previous 12 months been obliged or entitled (as applicable) to pay or receive compensation relating to the provision of services set out in Sections A and B of Annex I of Directive 2014/65/EU.
26. Within the preceding 12 months, Deutsche Bank and/or its affiliate(s) has received compensation for the provision of investment banking services or is currently providing or has provided investment banking services to this company.

For disclosures pertaining to recommendations or estimates made on securities other than the primary subject of this research, please see the most recently published company report or visit our global disclosure look-up page on our website at <https://research.db.com/Research/Disclosures/EquityResearchDisclosures>. Aside from within this report, important risk and conflict disclosures can also be found at <https://research.db.com/Research/Disclosures/Disclaimer>. Investors are strongly encouraged to review this information before investing.

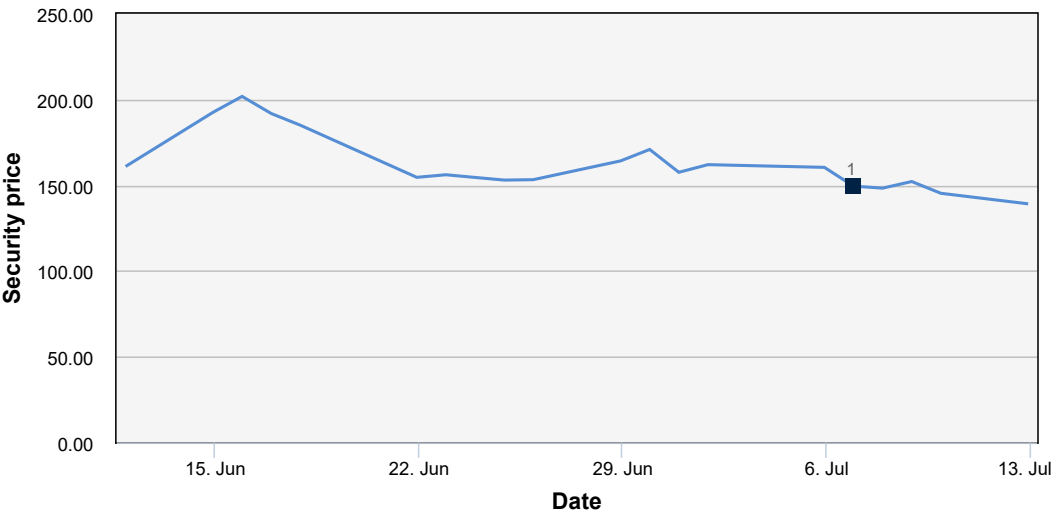
Analyst Certification

The views expressed in this report accurately reflect the personal views of the undersigned lead analyst(s) about the subject issuer and the securities of the issuer. In addition, the undersigned lead analyst(s) has not and will not receive any compensation for providing a specific recommendation or view in this report. Edison Yu.



Historical recommendations and target price: SpaceX (SPCX.OQ)

(as of 07/13/2026)



Current Recommendations

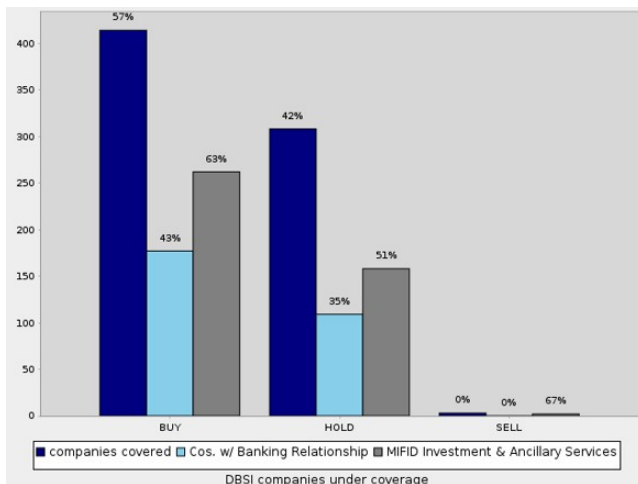
- Buy
- Hold
- Sell
- Not Rated
- Suspended Rating

** Analyst is no longer at Deutsche Bank

1. 07/07/2026 Buy, Target Price Change USD 255.00, Current Price USD 149.47 Edison Yu



Equity rating dispersion and banking relationships



Equity Rating and Dispersion Key

The Equity Rating Dispersion Chart depicts the following:

The proportion of recommendations that are rated "buy", "sell" and "hold" over the previous 12 months. This is shown for securities issued in the stated region e.g. "Europe Universe". See rating definitions below. This is represented by the "Companies Covered" bars in the chart. The percentage value displayed above the bar is the proportion as a percentage. E.g. 50% above the "buy" / "Companies Covered" bar means that 50% of DB's equity research covered companies over the past 12 months have a "buy" rating.

Next to each of the three respective bars showing the proportion of "buy", "sell" and "hold" recommendations we provide two additional bars to show:

- The proportion of "buy", "sell" or "hold" recommendations where Deutsche Bank and or/Affiliates provided MIFID Investment or Ancillary Services in the past 12 months. This is represented in the "MIFID Investment and Ancillary Services" bar. The percentage value displayed above the bar shows the proportion of Companies Covered with the given rating where DB has also provided MIFID Investment and Ancillary Services in the past 12 months. E.g. 50% above the "Cos. w/ MIFID Investment and Ancillary Services" bar means 50% of the Companies Covered with the rating stated have also received MIFID Investment and Ancillary Services from DB.

- The proportion of "buy" (or "sell" or "hold") recommendations where Deutsche Bank and or/Affiliates has provided Investment Banking services in the past 12 months for which it has received compensation. The percentage value displayed above the bar shows the proportion of Companies Covered with the stated rating where DB has also provided Investment Banking services in the past 12 months. E.g. 50% above the "Cos. w/ Investment Banking relationship" bar means 50% of the Companies Covered with the rating stated also have an Investment Banking Relationship with DB.

Buy: Based on a current 12- month view of TSR, we recommend that investors buy the stock.

Sell: Based on a current 12-month view of TSR, we recommend that investors sell the stock.

Hold: We take a neutral view on the stock 12-months out and, based on this time horizon, do not recommend either a Buy or Sell.

TSR = Total Shareholder Return. Percentage change in share price from current price to projected target price plus projected dividend yield

Newly issued research recommendations and target prices supersede previously published research.



Additional Information

The information and opinions in this report were prepared by Deutsche Bank AG or one of its affiliates (collectively 'Deutsche Bank'). Though the information herein is believed to be reliable and has been obtained from public sources believed to be reliable, Deutsche Bank makes no representation as to its accuracy or completeness. Hyperlinks to third-party websites in this report are provided for reader convenience only. Deutsche Bank neither endorses the content nor is responsible for the accuracy or security controls of those websites.

If you use the services of Deutsche Bank in connection with a purchase or sale of a security that is discussed in this report, or is included or discussed in another communication (oral or written) from a Deutsche Bank analyst, Deutsche Bank may act as principal for its own account or as agent for another person.

Deutsche Bank may consider this report in deciding to trade as principal. It may also engage in transactions, for its own account or with customers, in a manner inconsistent with the views taken in this research report. Others within Deutsche Bank, including strategists, sales staff and other analysts, may take views that are inconsistent with those taken in this research report. Deutsche Bank issues a variety of research products, including fundamental analysis, equity-linked analysis, quantitative analysis and trade ideas. Recommendations contained in one type of communication may differ from recommendations contained in others, whether as a result of differing time horizons, methodologies, perspectives or otherwise. Deutsche Bank and/or its affiliates may also be holding debt or equity securities of the issuers it writes on. Analysts are paid in part based on the profitability of Deutsche Bank AG and its affiliates, which includes investment banking, trading and principal trading revenues.

Opinions, estimates and projections constitute the current judgment of the author as of the date of this report. They do not necessarily reflect the opinions of Deutsche Bank and are subject to change without notice. Deutsche Bank provides liquidity for buyers and sellers of securities issued by the companies it covers. Deutsche Bank research analysts sometimes have shorter-term trade ideas that may be inconsistent with Deutsche Bank's existing longer-term ratings. Some trade ideas for equities are listed as Catalyst Calls on the Research Website (<https://research.db.com/Research/>), and can be found on the general coverage list and also on the covered company's page. A Catalyst Call represents a high-conviction belief by an analyst that a stock will outperform or underperform the market and/or a specified sector over a time frame of no less than two weeks and no more than three months. In addition to Catalyst Calls, analysts may occasionally discuss with our clients, and with Deutsche Bank salespersons and traders, trading strategies or ideas that reference catalysts or events that may have a near-term or medium-term impact on the market price of the securities discussed in this report, which impact may be directionally counter to the analysts' current 12-month view of total return or investment return as described herein. Deutsche Bank has no obligation to update, modify or amend this report or to otherwise notify a recipient thereof if an opinion, forecast or estimate changes or becomes inaccurate. Coverage and the frequency of changes in market conditions and in both general and company-specific economic prospects make it difficult to update research at defined intervals. Updates are at the sole discretion of the coverage analyst or of the Research Department Management, and the majority of reports are published at irregular intervals. This report is provided for informational purposes only and does not take into account the particular investment objectives, financial situations, or needs of individual clients. It is not an offer or a solicitation of an offer to buy or sell any financial instruments or to participate in any particular trading strategy. Target prices are inherently imprecise and a product of the analyst's judgment. The financial instruments discussed in this report may not be suitable for all investors, and investors must make their own informed investment decisions. Prices and availability of financial instruments are subject to change without notice, and investment transactions can lead to losses as a result of price fluctuations and other factors. If a financial instrument is denominated in a currency other than an investor's currency, a change in exchange rates may adversely affect the investment. Past performance is not necessarily indicative of future results. Performance calculations exclude transaction costs, unless otherwise indicated. Unless otherwise indicated, prices are current as of the end of the previous trading session and are sourced from local exchanges via Reuters, Bloomberg and other vendors. Data is also sourced from Deutsche Bank, subject companies, and other parties. Artificial intelligence tools may be used in the preparation of this material, including but not limited to assist in fact-finding, data analysis, pattern recognition, content drafting and editorial corrections pertaining to research material.

The Deutsche Bank Research Department is independent of other business divisions of the Bank. Details regarding our organizational arrangements and information barriers we have to prevent and avoid conflicts of interest with respect to our research are available on our website (<https://research.db.com/Research/>) under Disclaimer.

Macroeconomic fluctuations often account for most of the risks associated with exposures to instruments that promise to pay fixed or variable interest rates. For an investor who is long fixed-rate instruments (thus receiving these cash flows), increases in interest rates naturally lift the discount factors applied to the expected cash flows and thus cause a loss. The longer the maturity of a certain cash flow and the higher the move in the discount factor, the higher will be the loss. Upside surprises in inflation, fiscal funding needs, and FX depreciation rates are among the most common adverse macroeconomic shocks to receivers. But counterparty exposure, issuer creditworthiness, client segmentation, regulation (including changes in assets holding limits for different types of investors), changes in tax policies, currency convertibility (which may constrain currency conversion, repatriation of profits and/or liquidation of positions), and settlement issues related to local clearing houses are also important risk factors. The sensitivity of fixed-income instruments to macroeconomic shocks may be mitigated by indexing the contracted cash flows to inflation, to FX depreciation, or to specified interest rates - these are common in emerging markets. The index fixings may - by construction - lag or mis-measure the actual move in the underlying variables they are intended to track. The choice of the proper fixing (or metric) is particularly important in swaps markets, where floating coupon rates (i.e., coupons indexed to a typically short-dated interest rate reference index) are exchanged for fixed coupons. Funding in a currency that differs from the currency in which coupons are denominated carries FX risk. Options on swaps (swaptions) the risks typical to options in addition to the risks related to rates movements.



Derivative transactions involve numerous risks including market, counterparty default and illiquidity risk. The appropriateness of these products for use by investors depends on the investors' own circumstances, including their tax position, their regulatory environment and the nature of their other assets and liabilities; as such, investors should take expert legal and financial advice before entering into any transaction similar to or inspired by the contents of this publication. The risk of loss in futures trading and options, foreign or domestic, can be substantial. As a result of the high degree of leverage obtainable in futures and options trading, losses may be incurred that are greater than the amount of funds initially deposited - up to theoretically unlimited losses. Trading in options involves risk and is not suitable for all investors. Prior to buying or selling an option, investors must review the 'Characteristics and Risks of Standardized Options', at <https://www.theocc.com/company-information/documents-and-archives/options-disclosure-document>. If you are unable to access the website, please contact your Deutsche Bank representative for a copy of this important document.

Participants in foreign exchange transactions may incur risks arising from several factors, including the following: (i) exchange rates can be volatile and are subject to large fluctuations; (ii) the value of currencies may be affected by numerous market factors, including world and national economic, political and regulatory events, events in equity and debt markets and changes in interest rates; and (iii) currencies may be subject to devaluation or government-imposed exchange controls, which could affect the value of the currency. Investors in securities such as ADRs, whose values are affected by the currency of an underlying security, effectively assume currency risk.

Unless governing law provides otherwise, all transactions should be executed through the Deutsche Bank entity in the investor's home jurisdiction. Aside from within this report, important conflict disclosures can also be found at <https://research.db.com/Research/> on each company's research page or under the 'Disclosures' tab. Investors are strongly encouraged to review this information before investing.

Deutsche Bank (which includes Deutsche Bank AG, its branches and affiliated companies) is not acting as a financial adviser, consultant or fiduciary to you or any of your agents (collectively, "You" or "Your") with respect to any information provided in this report. Deutsche Bank does not provide investment, legal, tax or accounting advice, Deutsche Bank is not acting as your impartial adviser, and does not express any opinion or recommendation whatsoever as to any strategies, products or any other information presented in the materials. Information contained herein is being provided solely on the basis that the recipient will make an independent assessment of the merits of any investment decision, and it does not constitute a recommendation of, or express an opinion on, any product or service or any trading strategy.

The information presented is general in nature and is not directed to retirement accounts or any specific person or account type, and is therefore provided to You on the express basis that it is not advice, and You may not rely upon it in making Your decision. The information we provide is being directed only to persons we believe to be financially sophisticated, who are capable of evaluating investment risks independently, both in general and with regard to particular transactions and investment strategies, and who understand that Deutsche Bank has financial interests in the offering of its products and services. If this is not the case, or if You are an IRA or other retail investor receiving this directly from us, we ask that you inform us immediately.

In July 2018, Deutsche Bank revised its rating system for short term ideas whereby the branding has been changed to Catalyst Calls ("CC") from SOLAR ideas; the rating categories for Catalyst Calls originated in the Americas region have been made consistent with the categories used by Analysts globally; and the effective time period for CCs has been reduced from a maximum of 180 days to 90 days.

United States: Approved and/or distributed by Deutsche Bank Securities Incorporated, a member of FINRA and SIPC. Analysts located outside of the United States are employed by non-US affiliates and are not registered/qualified as research analysts with FINRA.

European Economic Area (exc. United Kingdom): Approved and/or distributed by Deutsche Bank AG, a joint stock corporation with limited liability incorporated in the Federal Republic of Germany with its principal office in Frankfurt am Main. Deutsche Bank AG is authorized under German Banking Law and is subject to supervision by the European Central Bank and by BaFin, Germany's Federal Financial Supervisory Authority.

United Kingdom: Approved and/or distributed by Deutsche Bank AG acting through its London Branch at 21 Moorfields, London EC2Y 9DB. Deutsche Bank AG in the United Kingdom is authorised by the Prudential Regulation Authority and is subject to limited regulation by the Prudential Regulation Authority and Financial Conduct Authority. Details about the extent of our authorisation and regulation are available on request.

Hong Kong SAR: Distributed by Deutsche Bank AG, Hong Kong Branch except for any research content relating to futures contracts within the meaning of the Hong Kong Securities and Futures Ordinance Cap. 571. Research reports on such futures contracts are not intended for access by persons who are located, incorporated, constituted or resident in Hong Kong. The author(s) of a research report, and the entities for which they act and/or are affiliated with, may not be licensed to carry on regulated activities in Hong Kong and, if not licensed, do not hold themselves out as being able to do so. The provisions set out above in the 'Additional Information' section shall apply to the fullest extent permissible by local laws and regulations, including without limitation the Code of Conduct for Persons Licensed or Registered with the Securities and Futures Commission. This report is intended for distribution only to 'professional investors' as defined in Part 1 of Schedule of the SFO. This document must not be acted or relied on by persons who are not professional investors. Any investment or investment activity to which this document relates is only available to professional investors and will be engaged only with professional investors.



India: Prepared by Deutsche Equities India Private Limited (DEIPL) having CIN: U65990MH2002PTC137431 and registered office at 14th Floor, The Capital, C-70, G Block, Bandra Kurla Complex, Mumbai (India) 400051. Tel: + 91 22 7180 4444. It is registered by the Securities and Exchange Board of India (SEBI) as a Stock broker bearing registration no.: INZ000252437; Merchant Banker bearing SEBI Registration no.: INM000010833 and Research Analyst bearing SEBI Registration no.: INH000001741. DEIPL's Compliance / Grievance officer is Ms. Rashmi Poddar (Tel: +91 22 7180 4929 email ID: complaints.deipl@db.com). Registration granted by SEBI and certification from NISM in no way guarantee performance of DEIPL or provide any assurance of returns to investors. Investment in securities market are subject to market risks. Read all the related documents carefully before investing. DEIPL may have received administrative warnings from the SEBI for breaches of Indian regulations. Deutsche Bank and/or its affiliate(s) may have debt holdings or positions in the subject company. With regard to information on associates, please refer to the "Shareholdings" section in the Annual Report at: <https://www.db.com/ir/en/annual-reports.htm>. For latest India research audit report, refer https://country.db.com/india/deutsche-equities-india/index?language_id=1.

Japan: Approved and/or distributed by Deutsche Securities Inc.(DSI). Registration number - Registered as a financial instruments dealer by the Head of the Kanto Local Finance Bureau (Kinsho) No. 117. Member of associations: JSDA, Type II Financial Instruments Firms Association and The Financial Futures Association of Japan. Commissions and risks involved in stock transactions - for stock transactions, we charge stock commissions and consumption tax by multiplying the transaction amount by the commission rate agreed with each customer. Stock transactions can lead to losses as a result of share price fluctuations and other factors. Transactions in foreign stocks can lead to additional losses stemming from foreign exchange fluctuations. We may also charge commissions and fees for certain categories of investment advice, products and services. Recommended investment strategies, products and services carry the risk of losses to principal and other losses as a result of changes in market and/or economic trends, and/or fluctuations in market value. Before deciding on the purchase of financial products and/or services, customers should carefully read the relevant disclosures, prospectuses and other documentation. 'Moody's', 'Standard Poor's', and 'Fitch' mentioned in this report are not registered credit rating agencies in Japan unless Japan or 'Nippon' is specifically designated in the name of the entity. Reports on Japanese listed companies not written by analysts of DSI are written by Deutsche Bank Group's analysts with the coverage companies specified by DSI. Some of the foreign securities stated on this report are not disclosed according to the Financial Instruments and Exchange Law of Japan. Target prices set by Deutsche Bank's equity analysts are based on a 12-month forecast period.

Korea: Distributed by Deutsche Securities Korea Co.

South Africa: Deutsche Bank AG Johannesburg is incorporated in the Federal Republic of Germany (Branch Register Number in South Africa: 1998/003298/10).

Singapore: This report is issued by Deutsche Bank AG, Singapore Branch (One Raffles Quay #18-00 South Tower Singapore 048583, 65 6423 8001), which may be contacted in respect of any matters arising from, or in connection with, this report. Where this report is issued or promulgated by Deutsche Bank in Singapore to a person who is not an accredited investor, expert investor or institutional investor (as defined in the applicable Singapore laws and regulations), they accept legal responsibility to such person for its contents.

Taiwan: Information on securities/investments that trade in Taiwan is for your reference only. Readers should independently evaluate investment risks and are solely responsible for their investment decisions. Deutsche Bank research may not be distributed to the Taiwan public media or quoted or used by the Taiwan public media without written consent. Information on securities/instruments that do not trade in Taiwan is for informational purposes only and is not to be construed as a recommendation to trade in such securities/instruments.

Qatar: Deutsche Bank AG in the Qatar Financial Centre (registered no. 00032) is regulated by the Qatar Financial Centre Regulatory Authority. Deutsche Bank AG - QFC Branch may undertake only the financial services activities that fall within the scope of its existing QFCRA license. Its principal place of business in the QFC: Qatar Financial Centre, Tower, West Bay, Level 5, PO Box 14928, Doha, Qatar. This information has been distributed by Deutsche Bank AG. Related financial products or services are only available only to Business Customers, as defined by the Qatar Financial Centre Regulatory Authority.

Russia: The information, interpretation and opinions submitted herein are not in the context of, and do not constitute, any appraisal or evaluation activity requiring a license in the Russian Federation.

Kingdom of Saudi Arabia: Deutsche Securities Saudi Arabia (DSSA) is a closed joint stock company authorized by the Capital Market Authority of the Kingdom of Saudi Arabia with a license number (No. 37-07073) to conduct the following business activities: Dealing, Arranging, Advising, and Custody activities. DSSA registered office is Faisaliah Tower, 17th Floor, King Fahad Road - Al Olaya District Riyadh, Kingdom of Saudi Arabia P.O. Box 301806.

United Arab Emirates: Deutsche Bank AG in the Dubai International Financial Centre (registered no. 00045) is regulated by the Dubai Financial Services Authority. Deutsche Bank AG - DIFC Branch may only undertake the financial services activities that fall within the scope of its existing DFSA license. Principal place of business in the DIFC: Dubai International Financial Centre, The Gate Village, Building 5, PO Box 504902, Dubai, U.A.E. This information has been distributed by Deutsche Bank AG. Related financial products or services are available only to Professional Clients, as defined by the Dubai Financial Services Authority.

Australia and New Zealand: This research is intended only for 'wholesale clients' within the meaning of the Australian Corporations Act and New Zealand Financial Advisors Act, respectively. Please refer to Australia-specific research disclosures



and related information at https://www.dbresearch.com/PROD/RPS_EN-PROD/PROD0000000000521304.xhtml. Where research refers to any particular financial product recipients of the research should consider any product disclosure statement, prospectus or other applicable disclosure document before making any decision about whether to acquire the product. In preparing this report, the primary analyst or an individual who assisted in the preparation of this report has likely been in contact with the company that is the subject of this research for confirmation/clarification of data, facts, statements, permission to use company-sourced material in the report, and/or site-visit attendance. Without prior approval from Research Management, analysts may not accept from current or potential Banking clients the costs of travel, accommodations, or other expenses incurred by analysts attending site visits, conferences, social events, and the like. Similarly, without prior approval from Research Management and Anti-Bribery and Corruption ("ABC") team, analysts may not accept perks or other items of value for their personal use from issuers they cover.

Additional information relative to securities, other financial products or issuers discussed in this report is available upon request. This report may not be reproduced, distributed or published without Deutsche Bank's prior written consent.

Backtested, hypothetical or simulated performance results have inherent limitations. Unlike an actual performance record based on trading actual client portfolios, simulated results are achieved by means of the retroactive application of a backtested model itself designed with the benefit of hindsight. Taking into account historical events the backtesting of performance also differs from actual account performance because an actual investment strategy may be adjusted any time, for any reason, including a response to material, economic or market factors. The backtested performance includes hypothetical results that do not reflect the reinvestment of dividends and other earnings or the deduction of advisory fees, brokerage or other commissions, and any other expenses that a client would have paid or actually paid. No representation is made that any trading strategy or account will or is likely to achieve profits or losses similar to those shown. Alternative modeling techniques or assumptions might produce significantly different results and prove to be more appropriate. Past hypothetical backtest results are neither an indicator nor guarantee of future returns. Actual results will vary, perhaps materially, from the analysis.

The method for computing individual E,S,G and composite ESG scores set forth herein is a novel method developed by the Research department within Deutsche Bank AG, computed using a systematic approach without human intervention. Different data providers, market sectors and geographies approach ESG analysis and incorporate the findings in a variety of ways. As such, the ESG scores referred to herein may differ from equivalent ratings developed and implemented by other ESG data providers in the market and may also differ from equivalent ratings developed and implemented by other divisions within the Deutsche Bank Group. Such ESG scores also differ from other ratings and rankings that have historically been applied in research reports published by Deutsche Bank AG. Further, such ESG scores do not represent a formal or official view of Deutsche Bank AG. It should be noted that the decision to incorporate ESG factors into any investment strategy may inhibit the ability to participate in certain investment opportunities that otherwise would be consistent with your investment objective and other principal investment strategies. The returns on a portfolio consisting primarily of sustainable investments may be lower or higher than portfolios where ESG factors, exclusions, or other sustainability issues are not considered, and the investment opportunities available to such portfolios may differ. Companies may not necessarily meet high performance standards on all aspects of ESG or sustainable investing issues; there is also no guarantee that any company will meet expectations in connection with corporate responsibility, sustainability, and/or impact performance.

Copyright © 2026 Deutsche Bank AG



David Folkerts-Landau

Group Chief Economist and Global Head of Research

Pam Finelli
COO and Head of Fixed
Income Research

Steve Pollard
Global Head of Company
Research and Sales

Jim Reid
Global Head of Macro and
Thematic Research

Tim Rokossa
Head of European
Company Research

Matthew Barnard
Head of Americas
Company Research

Debbie Jones
Global Head of
Sustainability and Data
Innovation, Research

Robin Winkler
Head of German Macro
Research

Sameer Goel
Global Head of EM &
APAC Research

Francis Yared
Global Head of Rates
Research

George Saravelos
Global Head of FX
Research

Peter Hooper
Vice-Chair of Research

Nilendra de-Mel
Head of APAC & Middle
East Product
Development

International Production Locations

Deutsche Bank AG

Deutsche Bank Place
Level 16
Corner of Hunter & Phillip
Streets
Sydney, NSW 2000
Australia
Tel: (61) 2 8258 1234

Deutsche Bank AG

Equity Research
Mainzer Landstrasse 11-17
60329 Frankfurt am Main
Germany
Tel: (49) 69 910 00

Deutsche Bank AG

Filiale Hongkong
International Commerce
Centre,
1 Austin Road West, Kowloon,
Hong Kong
Tel: (852) 2203 8888

Deutsche Securities Inc.

1-3-1 Azabudai
Azabudai Hills Mori JP Tower
Minato-ku, Tokyo 106-0041
Japan
Tel: (81) 3 6730 1000

Deutsche Bank AG

21 Moorfields
London EC2Y 9DB
United Kingdom
Tel: (44) 20 7545 8000

Deutsche Bank Securities Inc.

The Deutsche Bank Center
1 Columbus Circle
New York, NY 10019
Tel: (1) 212 250 2500

Deutsche Bank AG

Filiale Singapur
One Raffles Quay, South
Tower,
Singapore 048583
Tel: +65 6423 8001
